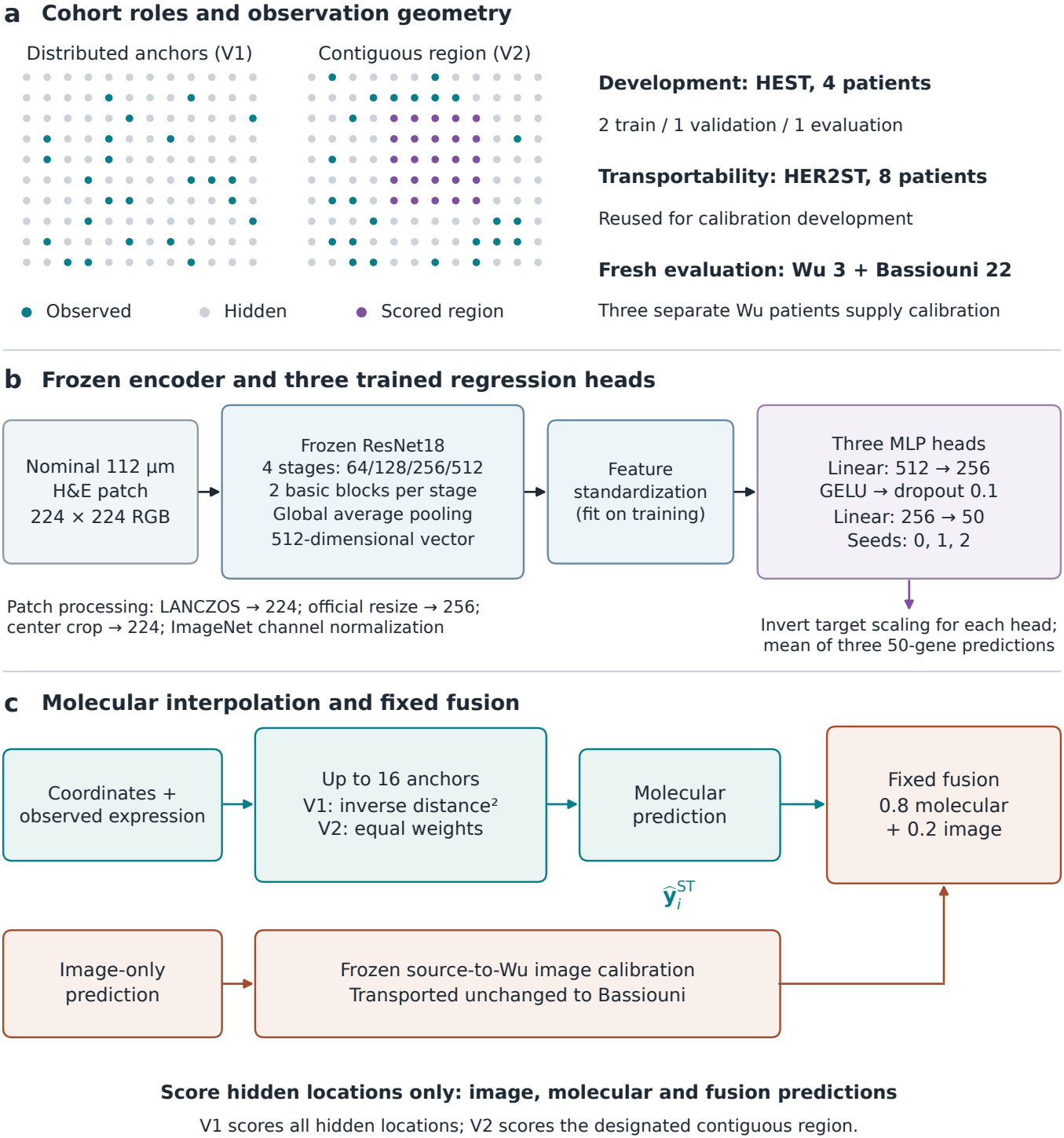
a Cohort roles and observation geometry
Distributed anchors (V1)
Contiguous region (V2)
Observed
Hidden
Scored region
Development: HEST, 4 patients
2 train / 1 validation / 1 evaluation
Transportability: HER2ST, 8 patients
Reused for calibration development
Fresh evaluation: Wu 3 + Bassiouni 22
Three separate Wu patients supply calibration
b Frozen encoder and three trained regression heads
Nominal 112 µm H&E patch 224 × 224 RGB
Frozen ResNet18 4 stages: 64/128/256/512 2 basic blocks per stage Global average pooling 512-dimensional vector
Feature standardization (fit on training)
Three MLP heads Linear: 512 → 256 GELU → dropout 0.1 Linear: 256 → 50 Seeds: 0, 1, 2
Patch processing: LANCZOS → 224; official resize → 256; center crop → 224; ImageNet channel normalization
Invert target scaling for each head; mean of three 50-gene predictions
c Molecular interpolation and fixed fusion
Coordinates + observed expression
Up to 16 anchors V1: inverse distance² V2: equal weights
Molecular prediction
$\hat{\mathbf{y}}_i^{ST}$
Fixed fusion 0.8 molecular + 0.2 image
Image-only prediction
Frozen source-to-Wu image calibration Transported unchanged to Bassiouni
Score hidden locations only: image, molecular and fusion predictions
V1 scores all hidden locations; V2 scores the designated contiguous region.